

Let p be a prime number.

Lemma. $\Phi_p(x) = \frac{x^p - 1}{x - 1} = x^{p-1} + x^{p-2} + \dots + x + 1$

is irreducible.

Proof. observe that $\Phi_p(x+1) = [(x+1)^p - 1] / x = \frac{1}{x} \cdot (x^p + \binom{p}{1}x^{p-1} + \dots + \binom{p}{p-1}x + \binom{p}{0} - 1)$
 $= x^{p-1} + \binom{p}{1}x^{p-2} + \dots + p,$

so applying the Eisenstein Criterion, we obtain the result.

Lemma. Let $\zeta_p \stackrel{\text{def}}{=} \exp(2\pi i \cdot \frac{1}{p})$. Then, the set

$$\{\zeta_p^0, \zeta_p^1, \dots, \zeta_p^{p-1}\} \cong \mathbb{Z}_p.$$

Moreover, $\mathbb{Q}(\zeta_p)$ is the splitting field of $x^p - 1$ and also Φ_p , with $[\mathbb{Q}(\zeta_p) : \mathbb{Q}] = p-1$.

Proof. It is trivial, since p is a prime number. $\square$

Lemma. $[\mathbb{Q}(\zeta_p + \zeta_p^{-1}) : \mathbb{Q}] = \frac{p-1}{2}$.

Proof. Since $\zeta_p^{-1} = \zeta_p^{p-1} = \exp(2\pi i \cdot \frac{p-1}{p})$, $\zeta_p + \zeta_p^{-1} = 2 \cdot \cos(2\pi \cdot \frac{1}{p}) \in \mathbb{R}$

Then, $x^2 - (\zeta_p + \zeta_p^{-1})x + 1 \in \mathbb{Q}(\zeta_p + \zeta_p^{-1})[x]$ and it has roots as ζ_p and ζ_p^{-1} .

For odd prime p ($p \geq 3$), $\zeta_p \notin \mathbb{Q}(\zeta_p + \zeta_p^{-1})$, so the polynomial above is irreducible.

Therefore, $[\mathbb{Q}(\zeta_p) : \mathbb{Q}(\zeta_p + \zeta_p^{-1})] = 2$. Now, the tower law gives the result.

(If $p=2$, then $\zeta_2 = -1$, so proved.)

Thm. Splitting field of x^{p-2} over $\mathbb{Q}$ is $\mathbb{Q}(\sqrt[p]{2}, \zeta_p)$, and $[\mathbb{Q}(\sqrt[p]{2}, \zeta_p) : \mathbb{Q}] = p(p-1)$.

Proof. Since for each $0 \leq i \leq p-1$, $(\sqrt[p]{2} \cdot \zeta_p^i)^p = 2 \cdot \zeta_p^{pi} = 2 \cdot 1 = 2$,

so the splitting field of x^{p-2} is $\mathbb{Q}(\sqrt[p]{2}, \zeta_p)$. To evaluate the degree, observe that

$$\mathbb{Q}(\sqrt[p]{2}, \zeta_p) = \mathbb{Q}(\sqrt[p]{2})\mathbb{Q}(\zeta_p) \quad \text{and}$$

$$[\mathbb{Q}(\sqrt[p]{2}) : \mathbb{Q}] = p \quad \text{and} \quad [\mathbb{Q}(\zeta_p) : \mathbb{Q}] = p-1, \text{ so}$$

$$[\mathbb{Q}(\sqrt[p]{2}, \zeta_p) : \mathbb{Q}] = p(p-1).$$

Example. $K = \mathbb{Q}(\sqrt{2}, \sqrt{3})$

The K is a splitting field for $(x^2-2) \cdot (x^2-3)$



But, the splitting field of x^3-2 is not just $\mathbb{Q}(\sqrt[3]{2})$,

because there are three roots:

$$\sqrt[3]{2}, \quad \sqrt[3]{2} \cdot \left(\frac{-1 + i\sqrt{3}}{2} \right), \quad \sqrt[3]{2} \cdot \left(\frac{-1 - i\sqrt{3}}{2} \right)$$

Let K be the splitting field of x^3-2 . Then is,

$$\sqrt[3]{2} \in K \quad \text{and} \quad \sqrt[3]{2} \cdot \left(\frac{-1 + i\sqrt{3}}{2} \right) \in K,$$

so $\sqrt{-3} \in K$. It follows that $K = \mathbb{Q}(\sqrt[3]{2}, \sqrt{-3})$.

Since $\mathbb{Q}(\sqrt[3]{2}, \sqrt{-3}) = \mathbb{Q}(\sqrt[3]{2}) \mathbb{Q}(\sqrt{-3})$ and

$[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$ and $[\mathbb{Q}(\sqrt{-3}) : \mathbb{Q}] = 2$, these are relatively prime,
thus the degree of composite field is 6.

Find the Galois group of $x^5 - 3 \in \mathbb{Q}(\zeta)[x]$ over $\mathbb{Q}(\zeta)$, where ζ is a primitive fifth root of unity.

Proof. Let K be the splitting field of $x^5 - 3$.

Since all roots of $x^5 - 3$ are

$$\sqrt[5]{3} \cdot \zeta^k \quad (k=0,1,2,3,4) \quad \text{where} \quad \zeta = \exp\left(2\pi i \cdot \frac{1}{5}\right) = e^{\frac{2\pi i}{5}},$$

Therefore $K = \mathbb{Q}(\sqrt[5]{3}, \zeta)$. Now,

$$K = \mathbb{Q}(\sqrt[5]{3}, \zeta)$$

$$\begin{array}{cc} 4 & \backslash & 5 \\ \mathbb{Q}(\sqrt[5]{3}) & & \mathbb{Q}(\zeta) \end{array}$$

$\begin{array}{cc} 5 & \backslash & 4 \\ \mathbb{Q} & & \mathbb{Q} \end{array}$ ($x^4 + x^3 + x^2 + x + 1$ has root ζ and is irreducible)

($\because x^5 - 3$ is irreducible over $\mathbb{Q}$)

Since $[\mathbb{Q}(\sqrt[5]{3}, \zeta) : \mathbb{Q}] = [\mathbb{Q}(\sqrt[5]{3}) : \mathbb{Q}] \cdot [\mathbb{Q}(\zeta) : \mathbb{Q}]$ and both degree over $\mathbb{Q}$ are relatively prime, $[K : \mathbb{Q}] = 20$. And, since $[K : \mathbb{Q}(\zeta)] = 5$, prime,

$$\text{Gal}(K/\mathbb{Q}(\zeta)) \cong \mathbb{Z}_5.$$